Area, Surface Area & Volume Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Area of a triangle

ORIGINAL PROBLEM

Find the area of a triangle with base 10 and height 6.

INCORRECT WORKED SOLUTION

$A = bh$

$A = 10 \cdot 6 = 60$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Area of a parallelogram

ORIGINAL PROBLEM

A parallelogram has base 8, slanted side 5 and height 4. Find its area.

INCORRECT WORKED SOLUTION

$A = \text{base} \cdot \text{side}$

$A = 8 \cdot 5 = 40$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Area of a trapezoid

ORIGINAL PROBLEM

Find the area of a trapezoid with bases 6 and 10 and height 4.

INCORRECT WORKED SOLUTION

$A = (b_1 + b_2)h$

$A = (6 + 10)(4) = 64$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Shaded regions

ORIGINAL PROBLEM

A circle with diameter 10 is cut out of a 10 by 10 square. Find the area of the shaded region.

INCORRECT WORKED SOLUTION

Square: $10 \cdot 10 = 100$

Circle: $\pi(10)^2 = 100\pi$

Shaded $= 100 - 100\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Volume of a cylinder

ORIGINAL PROBLEM

Find the volume of a cylinder with diameter 6 and height 7.

INCORRECT WORKED SOLUTION

$V = \pi r^2 h$

$V = \pi(6)^2(7)$

$V = 252\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Volume of a cone

ORIGINAL PROBLEM

Find the volume of a cone with radius 3 and height 4.

INCORRECT WORKED SOLUTION

$V = \pi r^2 h$

$V = \pi(3)^2(4)$

$V = 36\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Height vs. slant height

ORIGINAL PROBLEM

A cone has radius 6 and slant height 10. Find its volume.

INCORRECT WORKED SOLUTION

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(6)^2(10)$

$V = 120\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Volume of a sphere

ORIGINAL PROBLEM

Find the volume of a sphere with radius 3.

INCORRECT WORKED SOLUTION

$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi(3)^2$

$V = 12\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Area, Surface Area & Volume Error Analysis

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9. Surface area of a prism

ORIGINAL PROBLEM

Find the surface area of a rectangular prism that is 3 by 4 by 5.

INCORRECT WORKED SOLUTION

$2(3 \cdot 4) = 24$

$2(4 \cdot 5) = 40$

$SA = 24 + 40 = 64$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Square units

ORIGINAL PROBLEM

A rectangle is 2 ft by 3 ft. Find its area in square inches.

INCORRECT WORKED SOLUTION

Area $= 2 \cdot 3 = 6$ ft²

1 ft = 12 in

$6 \cdot 12 = 72$ in²

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points